

Plug Puller: the two-sided puller

Quick start, dial guide, measuring guide and measuring form.

Model version 0.13.0

In every picture: black = the tool; teal = your plug; red dotted = the edges this dial moved; the numbers match the key beside the picture.

The dial names are written exactly as the Customizer shows them. Use the bookmarks or the contents page to jump to a part or a dial.

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Measuring form

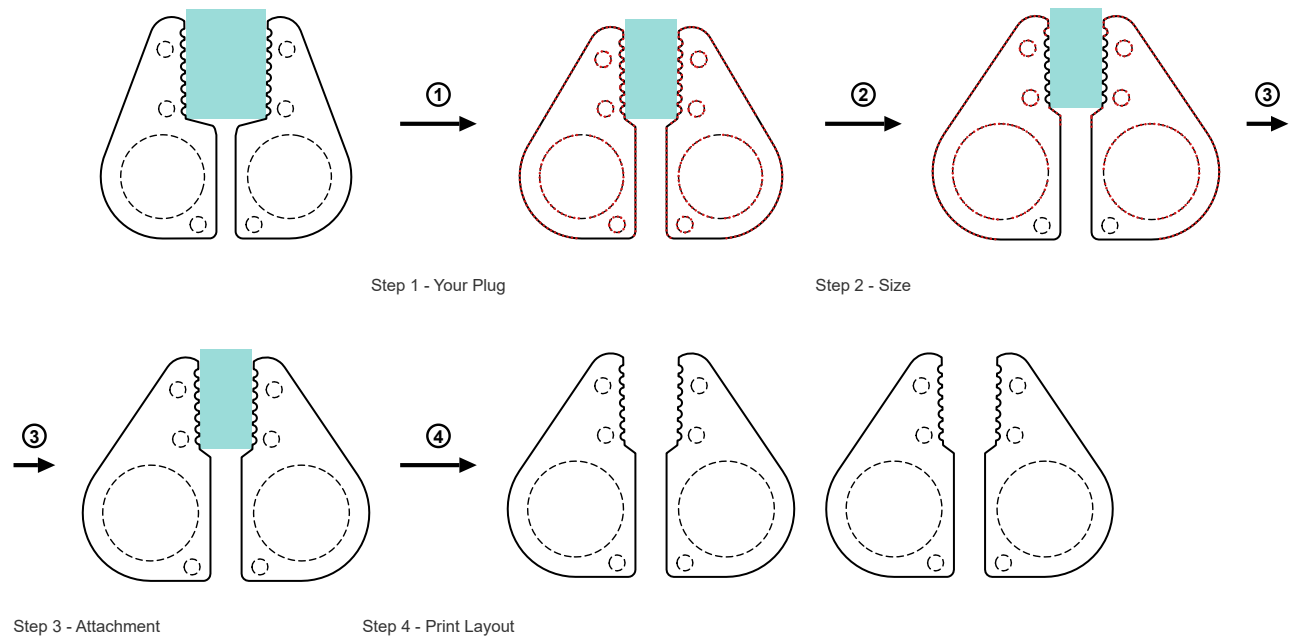
Which tool this is

This packet is for the two-sided puller, src/Plug_Puller_Two_Sided.scad: two serrated plates that zip-tie around the plug and close across it, for a plug 24 mm thick or more and for a plug held from both sides, a USB-C tip or a round extension-cord plug. A thinner wall plug is the one-sided puller's job, and that tool has its own packet.

The Customizer's four steps are your plug, your size, the attachment and the print layout. The quick start below shows each step's dials, the dial guide every dial of the file, the measuring guide how to take each number, and the measuring form is the sheet you fill in first.

If red text appears beside the part in the preview, read it: it names the measurement to fix, and the part will not fit until it is gone.

In every picture: black = the tool; teal = your plug; red dotted = the edges this dial moved; the numbers match the key beside the picture.



Five stages of the two-sided puller, left to right, the top row first, the plug end at the top; red dots mark the edges each step moved. 1, the defaults: the tool as the file opens, with a 20 mm wide plug in teal. 2, Step 1 - Your Plug: plug length 23 mm, plug width at the prong end 13 mm, plug width at the cord end 13 mm, cord thickness 7 mm, plug sides Rounded sides; red on the arms, the finger lobes and the zip stations. 3, Step 2 - Size: hand size Large; red on the arms, the finger lobes and the zip stations. 4, Step 3 - Attachment: attachment Zip ties; no strap slot on a plug this short, so nothing moved. 5, Step 4 - Print Layout: both plates side by side in one file, what you print; nothing marked.

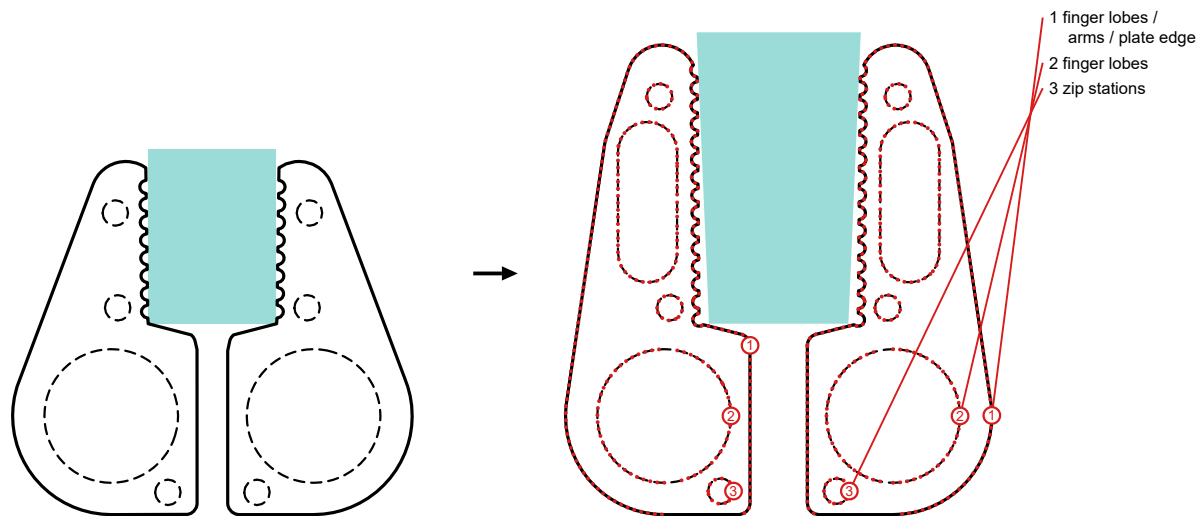
- Step 1: type your plug's numbers and pick Rounded sides or Flat sides
- Step 2: pick your hand size
- Step 3: pick how it attaches; a plug this short gets no strap slot
- Step 4: both plates in one file

Quick start

Step 1 - Your Plug

plug_preset

Plug preset



Measure my plug
→ Heavy-duty extension cord - NEMA 5-15

Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug preset set to Heavy-duty extension cord - NEMA 5-15. Marked in red: 1, the finger lobes, the arms and the plate edge. 2, the finger lobes: the two rounded lobes in the lower half. 3, the zip stations: the three small holes along each arm.

Fills in the plug's length, both widths and the cord from the measured heavy-duty cord plug, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.

Default Measure my plug · 5 options

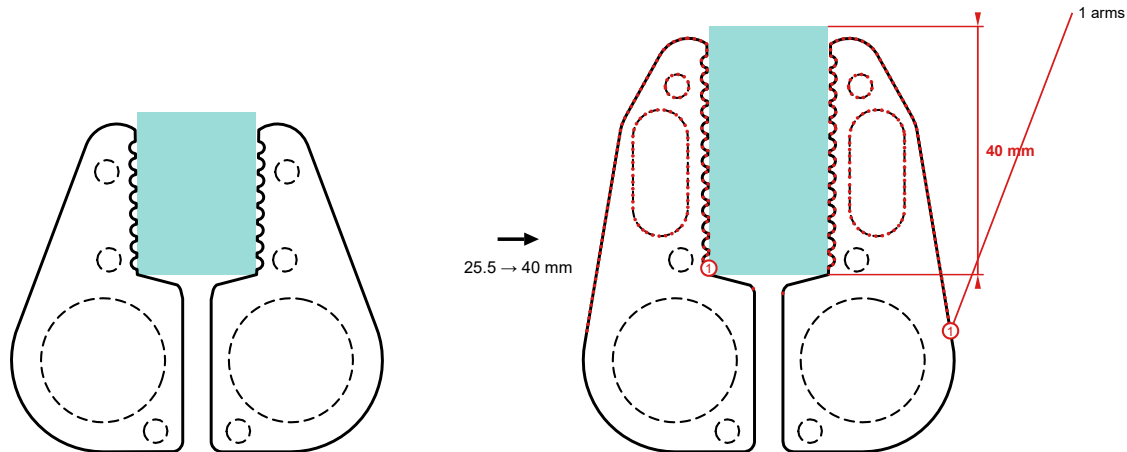
Options: Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15.

Before: Measure my plug. After: Heavy-duty extension cord - NEMA 5-15.

Moves: arms, finger lobes, plate edge, zip stations.

measure_plug_length

Plug length



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug length at 40 mm: lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them. Marked in red: 1, the arms: the two toothed arms in the upper half. The plate edge, the teeth, the finger lobes and the cord channel stay where they were.

Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.

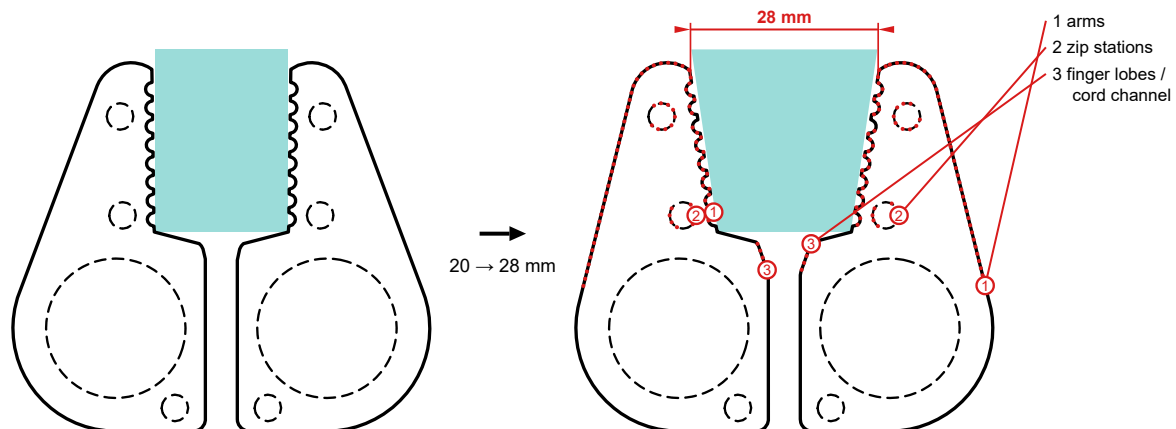
Default 25.5 · Range 12 to 85 · Step size 0.5 · Unit mm

Before: 25.5. After: 40.

Moves: arms.

measure_plug_width_prong_end

Plug width at the prong end



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the prong end at 28 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm. 3, the finger lobes and the cord channel: the two rounded lobes in the lower half; the gap between the lobes at the bottom.

Opens or closes the gap between the arms at their tips, where the plug's prong end sits.

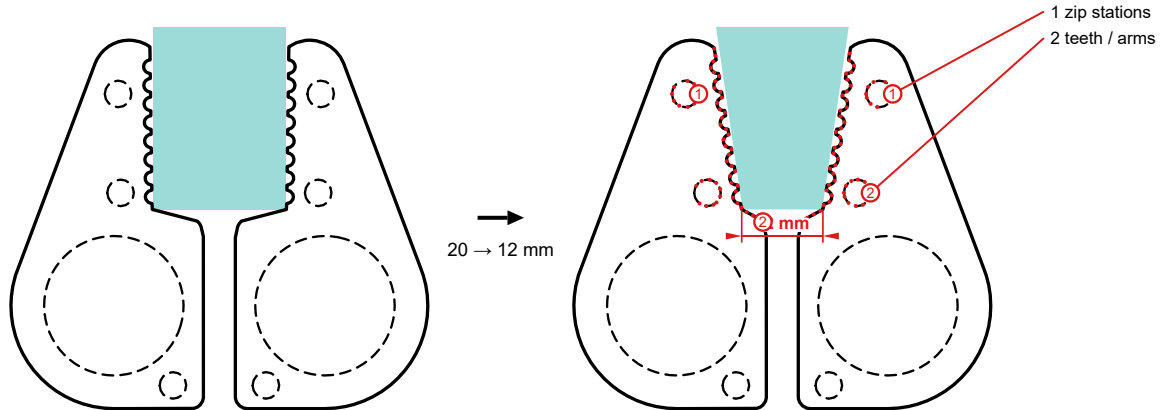
Default 20 · Range 5 to 40 · Step size 0.5 · Unit mm

Before: 20. After: 28.

Moves: arms, cord channel, finger lobes, zip stations.

measure_plug_width_cord_end

Plug width at the cord end



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the cord end at 12 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.

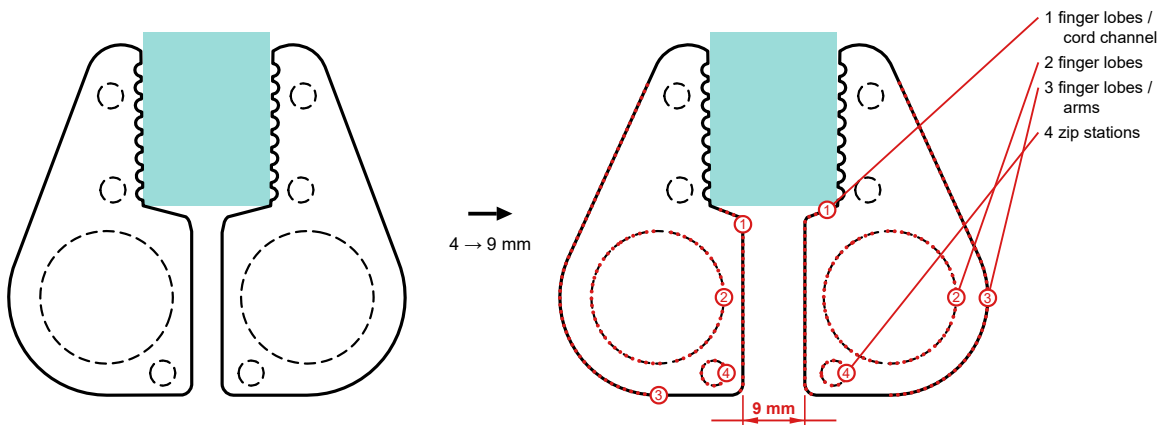
Default 20 · Range 5 to 40 · Step size 0.5 · Unit mm

Before: 20. After: 12.

Moves: arms, teeth, zip stations.

measure_cord_thickness

Cord thickness



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: cord thickness at 9 mm. Marked in red: 1, the finger lobes and the cord channel. 2, the finger lobes: the two rounded lobes in the lower half. 3, the finger lobes and the arms. 4, the zip stations: the three small holes along each arm.

Widens the cord channel between the finger holes, and the finger lobes move outward with it.

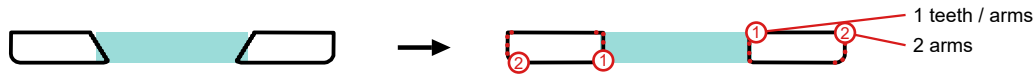
Default 4 · Range 1.5 to 12 · Step size 0.5 · Unit mm

Before: 4. After: 9.

Moves: arms, cord channel, finger lobes, zip stations.

plug_sides

Plug sides



Rounded sides
→ Flat sides
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug sides set to Flat sides. Marked in red: 1, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half. 2, the arms: the two toothed arms in the upper half.

Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.

Default Rounded sides · 2 options

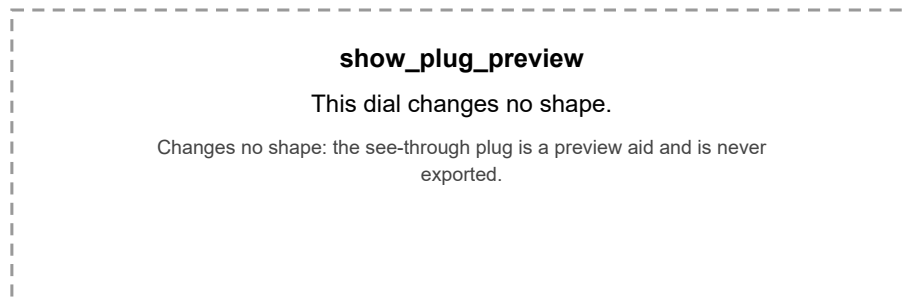
Options: Rounded sides, Flat sides.

Before: Rounded sides. After: Flat sides. Section at y = 45 mm.

Moves: arms, teeth.

show_plug_preview

Show the see-through plug



This dial changes no shape: the see-through plug is a preview aid and is never exported.

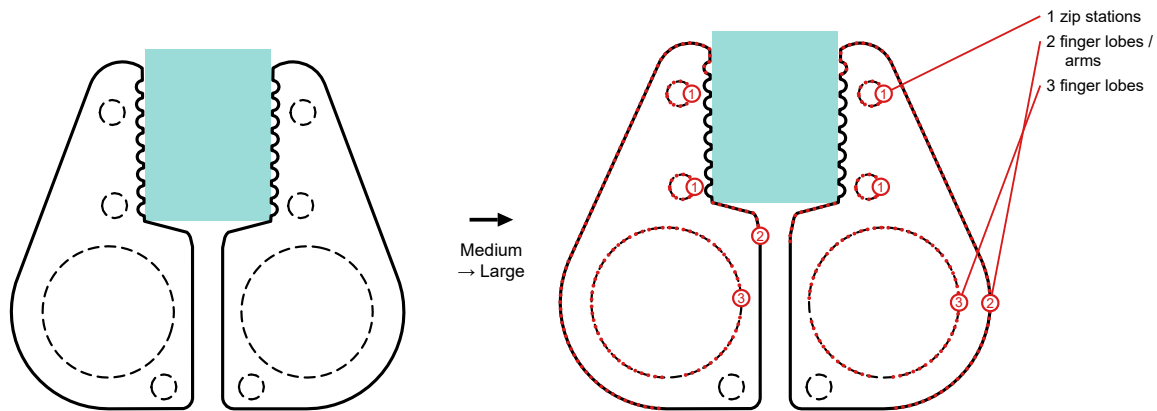
Shows or hides the see-through plug in the preview and changes nothing in the printed plates.

Default true · true / false

Step 2 - Size

size

Hand size



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: hand size set to Large. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Widens both finger holes and their lobes, so the whole plate grows around them.

Default Medium · 4 options

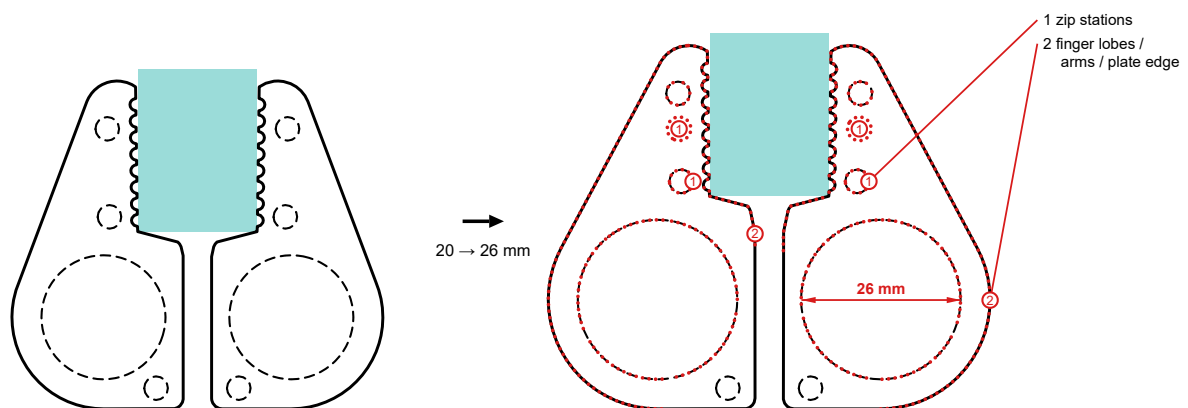
Options: Small, Medium, Large, Measure my hand.

Before: Medium. After: Large.

Moves: arms, finger lobes, zip stations.

measure_finger_width

Finger width



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 20 mm wide plug in teal. Right: finger width at 26 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes, the arms and the plate edge: the two rounded lobes in the lower half; the two toothed arms in the upper half; the outer outline.

Widens both finger holes and their lobes.

Default 20 · Range 14 to 32 · Step size 0.5 · Unit mm

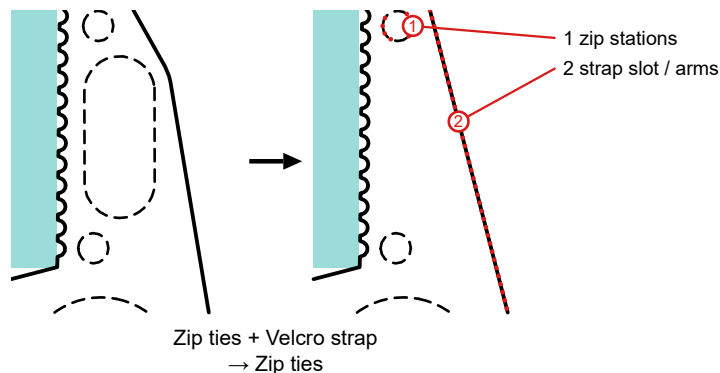
Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: arms, finger lobes, plate edge, zip stations.

Step 3 - Attachment

attachment Attachment



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: attachment set to Zip ties. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.

Default Zip ties + Velcro strap · 2 options

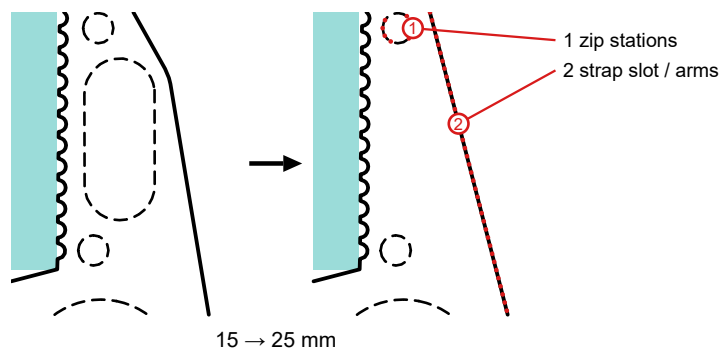
Options: Zip ties + Velcro strap, Zip ties.

Before: Zip ties + Velcro strap. After: Zip ties. Context: `measure_plug_length = 40`.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

strap_width Strap width



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap width at 25 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.

Default 15 · Range 10 to 25 · Step size 1 · Unit mm

Before: 15. After: 25. Context: `measure_plug_length = 40`.

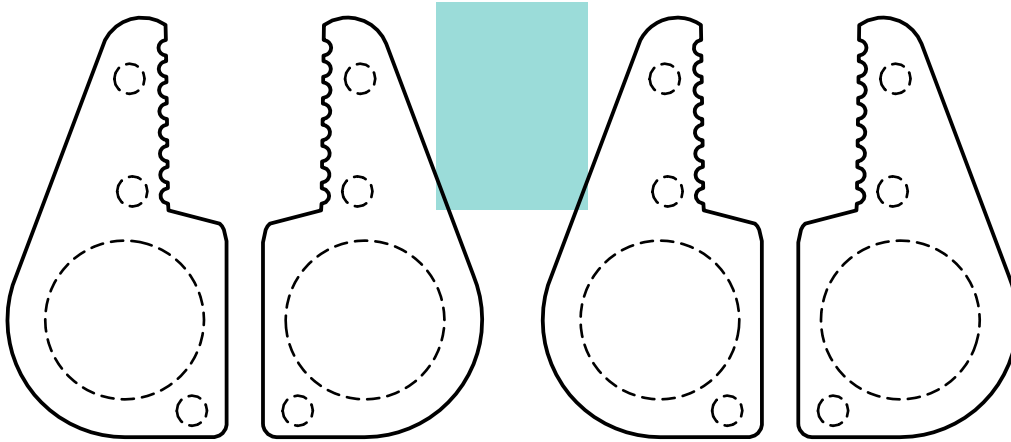
Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot. At 25 mm the slot leaves too, because the plug is too short for one, so the after picture has no slot and no dimension line.

Moves: arms, strap slot, zip stations.

Can trip: STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

Step 4 - Print Layout

print_layout Print layout



Both plates
→ One plate

One top view of the two-sided puller with both plates side by side, the plug end at the top: what the file prints at Both plates. At One plate it prints one plate. Nothing is marked in red: this dial changes the layout, not the plate.

Puts both identical plates side by side in one file, or just one plate.

Default Both plates · 2 options

Options: Both plates, One plate.

Before: Both plates. After: One plate.

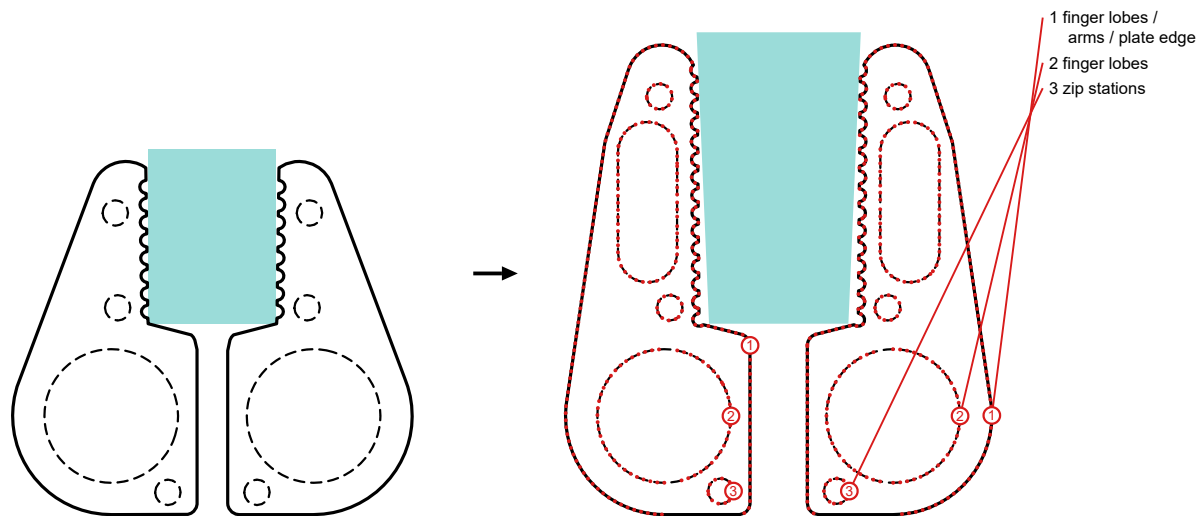
Moves: arms, finger lobes, plate edge, zip stations.

Full dial guide

Step 1 - Your Plug

plug_preset

Plug preset



Measure my plug
→ Heavy-duty extension cord - NEMA 5-15

Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug preset set to Heavy-duty extension cord - NEMA 5-15. Marked in red: 1, the finger lobes, the arms and the plate edge. 2, the finger lobes: the two rounded lobes in the lower half. 3, the zip stations: the three small holes along each arm.

Fills in the plug's length, both widths and the cord from the measured heavy-duty cord plug, so the arms, the grip gap, the cord channel and the plate length all take that plug's shape at once.

Default Measure my plug · 5 options

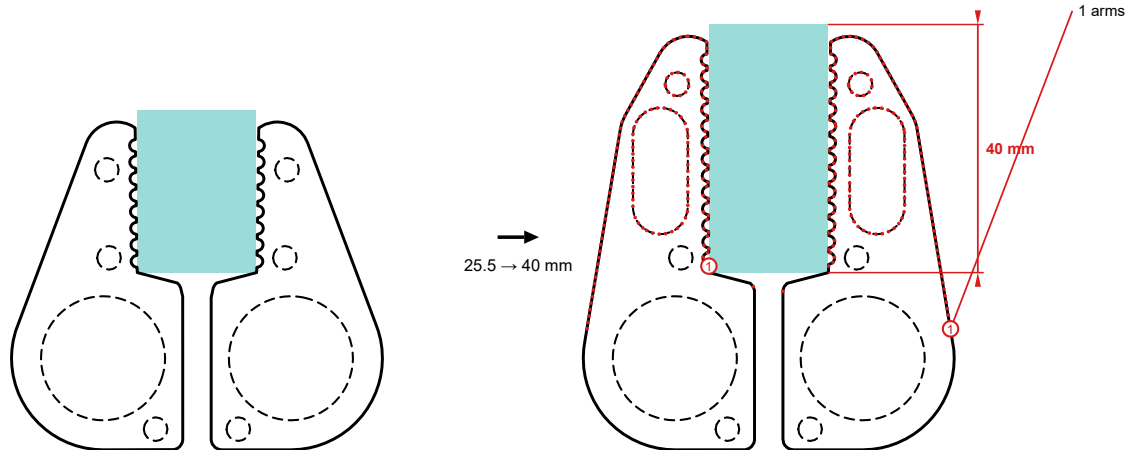
Options: Measure my plug, Heavy-duty extension cord - NEMA 5-15, USB-C laptop tip, Flat 2-prong lamp plug - NEMA 1-15, Standard 3-prong plug - NEMA 5-15.

Before: Measure my plug. After: Heavy-duty extension cord - NEMA 5-15.

Moves: arms, finger lobes, plate edge, zip stations.

measure_plug_length

Plug length



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug length at 40 mm: lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them. Marked in red: 1, the arms: the two toothed arms in the upper half. The plate edge, the teeth, the finger lobes and the cord channel stay where they were.

Lengthens both arms so the teeth cover the whole plug body, and moves the zip-tie stations and the strap slot with them.

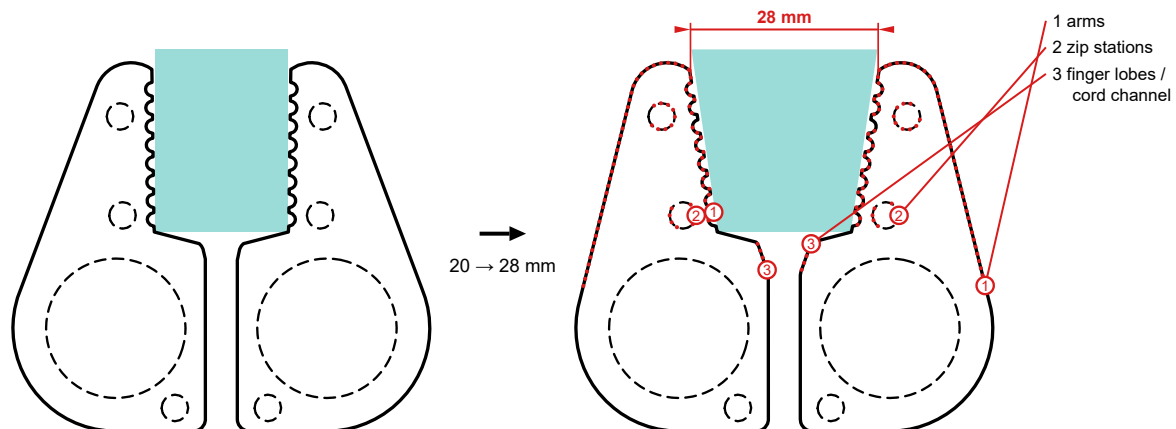
Default 25.5 · Range 12 to 85 · Step size 0.5 · Unit mm

Before: 25.5. After: 40.

Moves: arms.

measure_plug_width_prong_end

Plug width at the prong end



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the prong end at 28 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm. 3, the finger lobes and the cord channel: the two rounded lobes in the lower half; the gap between the lobes at the bottom.

Opens or closes the gap between the arms at their tips, where the plug's prong end sits.

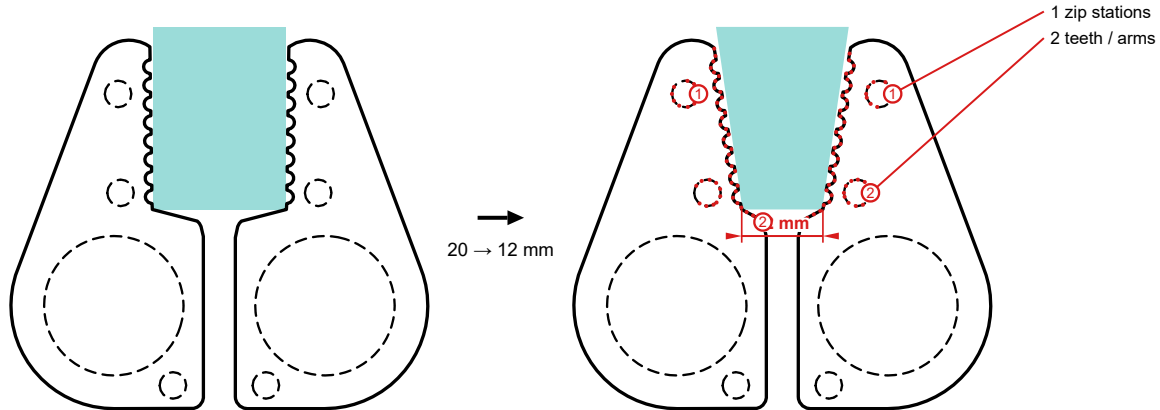
Default 20 · Range 5 to 40 · Step size 0.5 · Unit mm

Before: 20. After: 28.

Moves: arms, cord channel, finger lobes, zip stations.

measure_plug_width_cord_end

Plug width at the cord end



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: plug width at the cord end at 12 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Opens or closes the gap between the arms at the plug's back end, so the arms taper to match the plug.

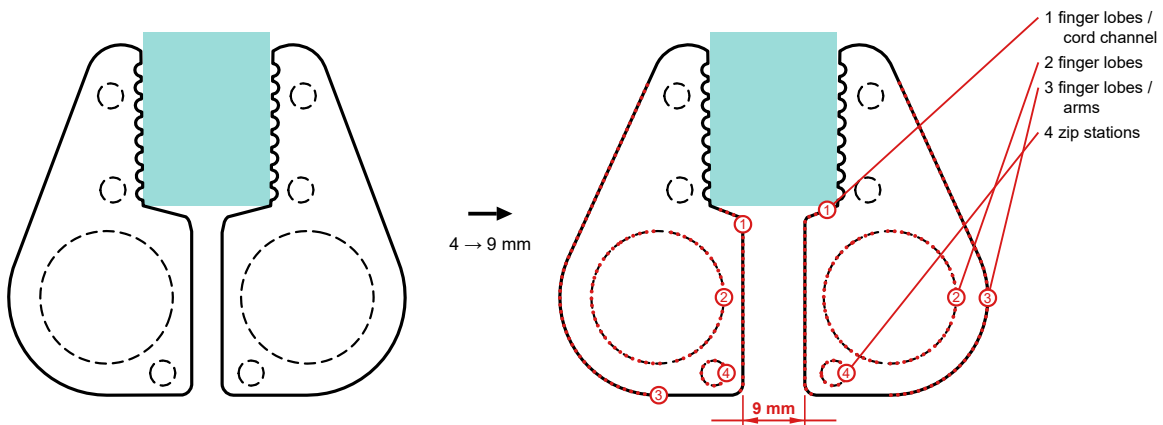
Default 20 · Range 5 to 40 · Step size 0.5 · Unit mm

Before: 20. After: 12.

Moves: arms, teeth, zip stations.

measure_cord_thickness

Cord thickness



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: cord thickness at 9 mm. Marked in red: 1, the finger lobes and the cord channel. 2, the finger lobes: the two rounded lobes in the lower half. 3, the finger lobes and the arms. 4, the zip stations: the three small holes along each arm.

Widens the cord channel between the finger holes, and the finger lobes move outward with it.

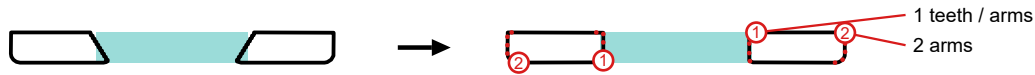
Default 4 · Range 1.5 to 12 · Step size 0.5 · Unit mm

Before: 4. After: 9.

Moves: arms, cord channel, finger lobes, zip stations.

plug_sides

Plug sides



Rounded sides
→ Flat sides
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plug sides set to Flat sides. Marked in red: 1, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half. 2, the arms: the two toothed arms in the upper half.

Removes the sloped cradle from the arms' gripping edges so the teeth bite straight along a flat-sided plug.

Default Rounded sides · 2 options

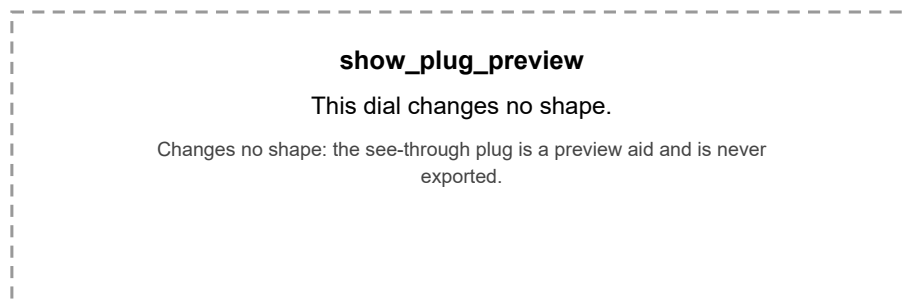
Options: Rounded sides, Flat sides.

Before: Rounded sides. After: Flat sides. Section at y = 45 mm.

Moves: arms, teeth.

show_plug_preview

Show the see-through plug



This dial changes no shape: the see-through plug is a preview aid and is never exported.

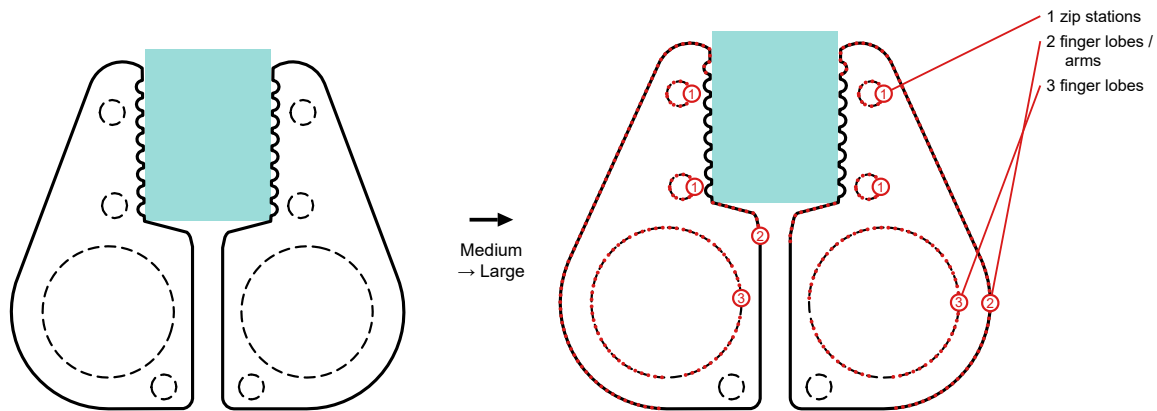
Shows or hides the see-through plug in the preview and changes nothing in the printed plates.

Default true · true / false

Step 2 - Size

size

Hand size



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: hand size set to Large. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Widens both finger holes and their lobes, so the whole plate grows around them.

Default Medium · 4 options

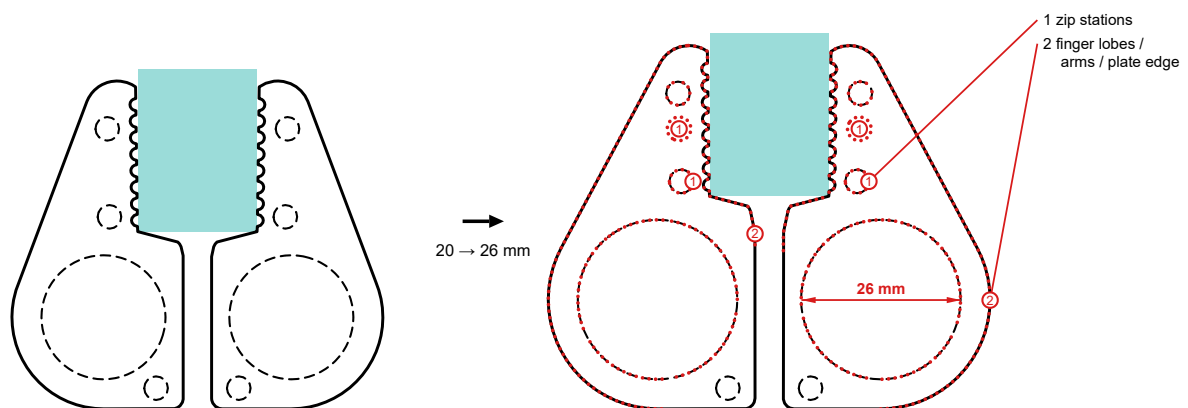
Options: Small, Medium, Large, Measure my hand.

Before: Medium. After: Large.

Moves: arms, finger lobes, zip stations.

measure_finger_width

Finger width



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with hand size set to Measure my hand, a 20 mm wide plug in teal. Right: finger width at 26 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes, the arms and the plate edge: the two rounded lobes in the lower half; the two toothed arms in the upper half; the outer outline.

Widens both finger holes and their lobes.

Default 20 · Range 14 to 32 · Step size 0.5 · Unit mm

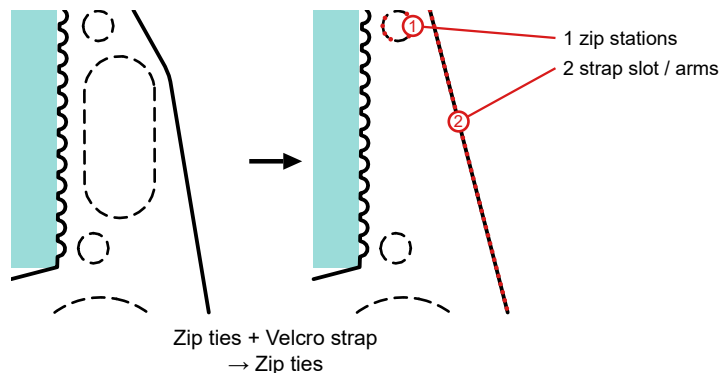
Before: 20. After: 26. Context: size = Measure my hand.

Only acts when size is Measure my hand.

Moves: arms, finger lobes, plate edge, zip stations.

Step 3 - Attachment

attachment Attachment



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: attachment set to Zip ties. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Removes the strap slot from each arm, while the zip-tie stations stay because they hold the plates together.

Default Zip ties + Velcro strap · 2 options

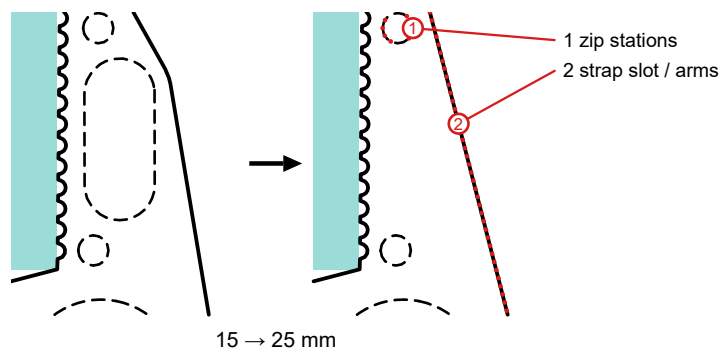
Options: Zip ties + Velcro strap, Zip ties.

Before: Zip ties + Velcro strap. After: Zip ties. Context: `measure_plug_length = 40`.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

strap_width Strap width



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap width at 25 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Sets the strap the arm slot must clear, and when the slot's window between the zip-tie stations is shorter than the strap plus 1.5 mm the slot is left out.

Default 15 · Range 10 to 25 · Step size 1 · Unit mm

Before: 15. After: 25. Context: `measure_plug_length = 40`.

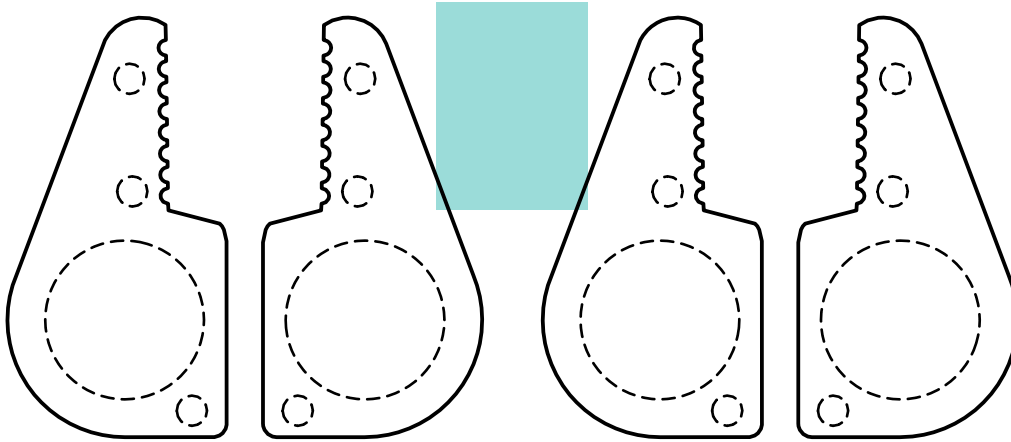
Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot. At 25 mm the slot leaves too, because the plug is too short for one, so the after picture has no slot and no dimension line.

Moves: arms, strap slot, zip stations.

Can trip: STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

Step 4 - Print Layout

print_layout Print layout



Both plates
→ One plate

One top view of the two-sided puller with both plates side by side, the plug end at the top: what the file prints at Both plates. At One plate it prints one plate. Nothing is marked in red: this dial changes the layout, not the plate.

Puts both identical plates side by side in one file, or just one plate.

Default Both plates · 2 options

Options: Both plates, One plate.

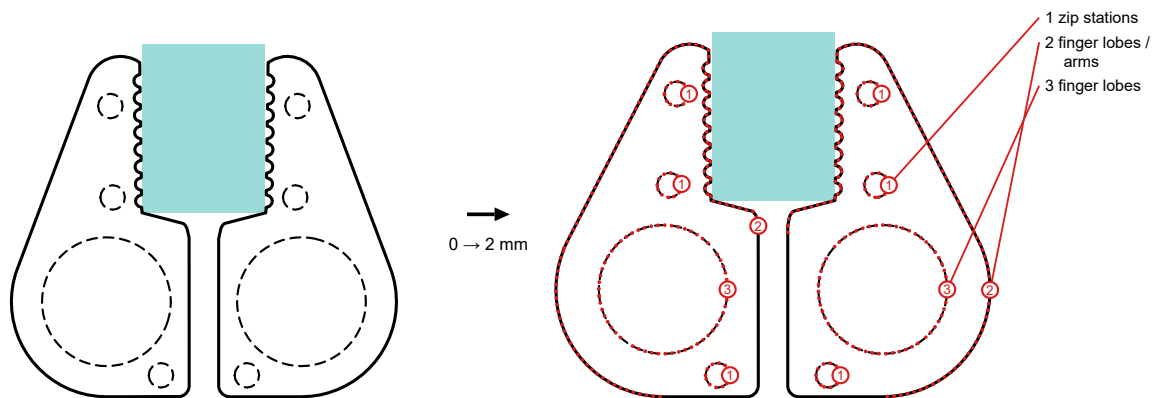
Before: Both plates. After: One plate.

Moves: arms, finger lobes, plate edge, zip stations.

Advanced - Two-Sided Puller

plate_wall_boost

Extra wall on the plate



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: extra wall on the plate at 2 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Thickens every wall around the finger holes, the cord channel, the zip-tie holes and the strap slot at once, so the plate outline grows and the openings shift to keep their walls.

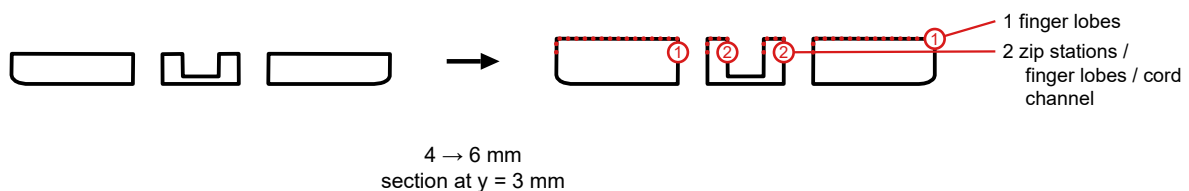
Default 0 · Range 0 to 5 · Step size 0.25 · Unit mm

Before: 0. After: 2.

Moves: arms, finger lobes, zip stations.

plate_thickness

Plate thickness



Two vertical slices of the two-sided puller at $y = 3$ mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: plate thickness at 6 mm. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. 2, the zip stations, the finger lobes and the cord channel: the three small holes along each arm; the two rounded lobes in the lower half; the gap between the lobes at the bottom.

Thickens each plate, so the finished pair is twice as thick.

Default 4 · Range 2 to 8 · Step size 0.25 · Unit mm

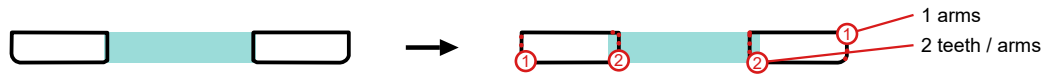
Before: 4. After: 6. Section at $y = 3$ mm.

Moves: cord channel, finger lobes, zip stations.

Can trip: PLATE THINNER THAN 2MM - TOO FLIMSY.

plate_grip_bite

Grip bite



-1 → -2 mm
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults with plug sides set to Flat sides, the plug in teal in the cut. Right: grip bite at -2 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Squeezes the plug harder or softer, because the arms close in by this much per side against the plug's width.

Default -1 · Range -2 to 2 · Step size 0.1 · Unit mm

Before: -1. After: -2. Context: plug_sides = Flat sides. Section at y = 45 mm.

Only acts when plug_sides is Flat sides or a plug preset is chosen.

Moves: arms, teeth.

Can trip: NO GRIP BITE - PLUG WONT BE HELD.

plate_cradle_depth

Cradle depth



2.5 → 0 mm
section at y = 45 mm

Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults with plug sides set to Rounded sides, the plug in teal in the cut. Right: cradle depth at 0 mm. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Slopes each arm's gripping edge from the mating face down to the outer face, so two plates form a cradle that centers a round plug.

Default 2.5 · Range 0 to 3.5 · Step size 0.25 · Unit mm

Before: 2.5. After: 0. Context: plug_sides = Rounded sides. Section at y = 45 mm.

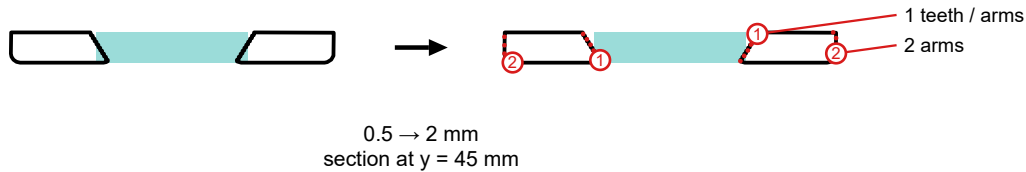
Only acts when plug_sides is Rounded sides.

Moves: arms, teeth.

Can trip: CRADLE SHALLOWER THAN ASKED - PLUG NARROW.

plate_grip_clearance

Grip clearance



Two vertical slices of the two-sided puller at y = 45 mm, before left and after right, the top face up. Left: the defaults with plug sides set to Rounded sides, the plug in teal in the cut. Right: grip clearance at 2 mm. Marked in red: 1, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half. 2, the arms: the two toothed arms in the upper half.

Opens the gap between the arms where the plates meet, so a hard round plug can drop in flat before the cradle holds it.

Default 0.5 · Range 0 to 2 · Step size 0.1 · Unit mm

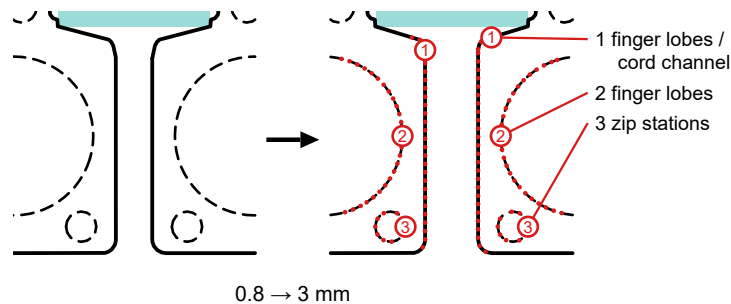
Before: 0.5. After: 2. Context: plug_sides = Rounded sides. Section at y = 45 mm.

Only acts when plug_sides is Rounded sides.

Moves: arms, teeth.

plate_cable_clearance

Cord channel clearance



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: cord channel clearance at 3 mm. Marked in red: 1, the finger lobes and the cord channel: the two rounded lobes in the lower half; the gap between the lobes at the bottom. 2, the finger lobes: the two rounded lobes in the lower half. 3, the zip stations: the three small holes along each arm.

Widens the cord channel between the finger holes beyond the cord's thickness.

Default 0.8 · Range 0 to 5 · Step size 0.1 · Unit mm

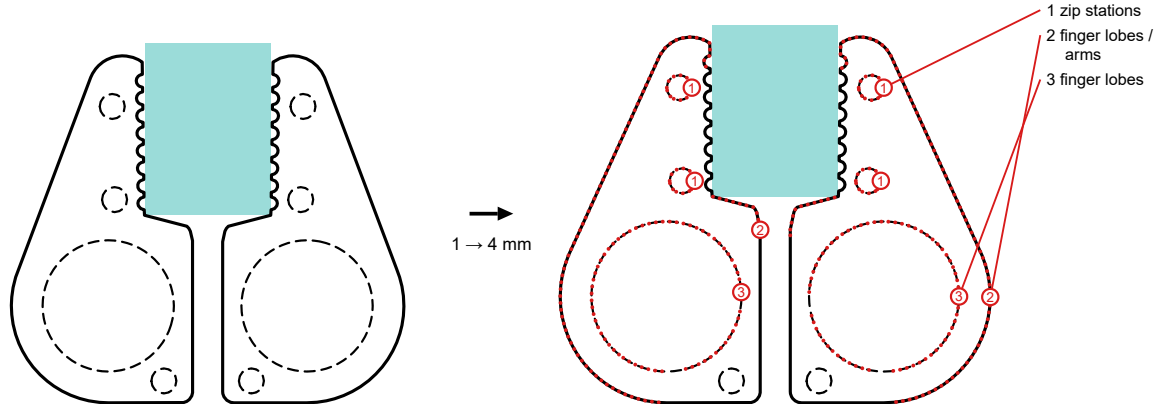
Before: 0.8. After: 3.

Moves: arms, cord channel, finger lobes, zip stations.

Can trip: CORD TOO THICK FOR CABLE CHANNEL; PLUG NARROWER THAN THE CORD CHANNEL - ARMS CANNOT TOUCH IT.

plate_finger_fit

Finger hole fit



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: finger hole fit at 4 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Widens both finger holes beyond your finger width.

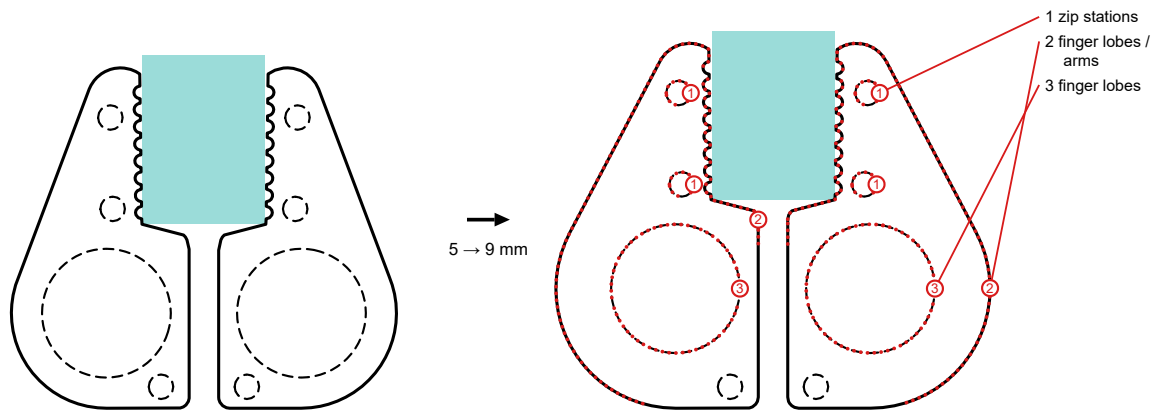
Default 1 · Range 0 to 8 · Step size 0.25 · Unit mm

Before: 1. After: 4.

Moves: arms, finger lobes, zip stations.

plate_finger_wall

Finger lobe wall



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: finger lobe wall at 9 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half. 3, the finger lobes: the two rounded lobes in the lower half.

Thickens the wall around each finger hole, so the rounded lobes grow.

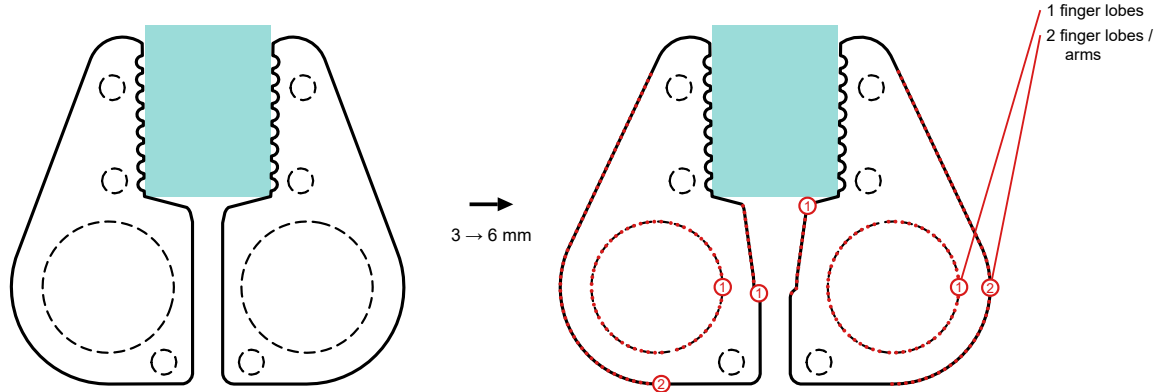
Default 5 · Range 3 to 12 · Step size 0.25 · Unit mm

Before: 5. After: 9.

Moves: arms, finger lobes, zip stations.

plate_finger_inner_wall

Wall beside the cord channel



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: wall beside the cord channel at 6 mm. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. 2, the finger lobes and the arms: the two rounded lobes in the lower half; the two toothed arms in the upper half.

Thickens the wall between the cord channel and each finger hole, pushing the finger holes outward.

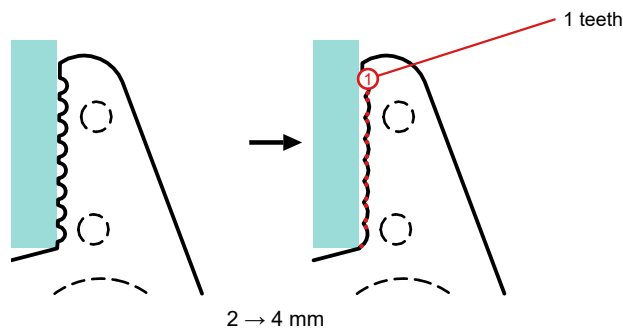
Default 3 · Range 1 to 8 · Step size 0.25 · Unit mm

Before: 3. After: 6.

Moves: arms, finger lobes, zip stations.

plate_tooth_diameter

Tooth size



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tooth size at 4 mm: cuts bigger or smaller scallops into the arms' gripping edges. Marked in red: 1, the teeth: the serrated inner edges of the arms. The plate edge, the arms, the finger lobes and the cord channel stay where they were.

Cuts bigger or smaller scallops into the arms' gripping edges.

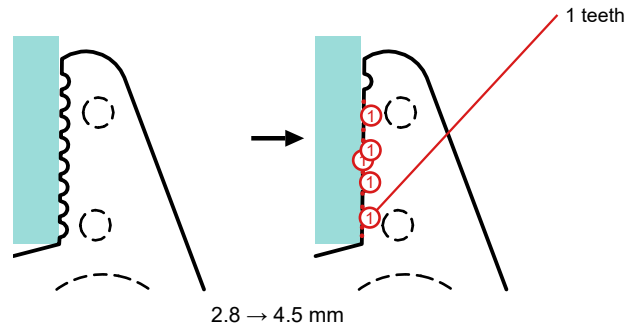
Default 2 · Range 0 to 4 · Step size 0.1 · Unit mm

Before: 2. After: 4.

Moves: teeth.

plate_tooth_pitch

Tooth spacing



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tooth spacing at 4.5 mm: spaces the teeth farther apart or closer together along the gripping edges. Marked in red: 1, the teeth: the serrated inner edges of the arms. The plate edge, the arms, the finger lobes and the cord channel stay where they were.

Spaces the teeth farther apart or closer together along the gripping edges.

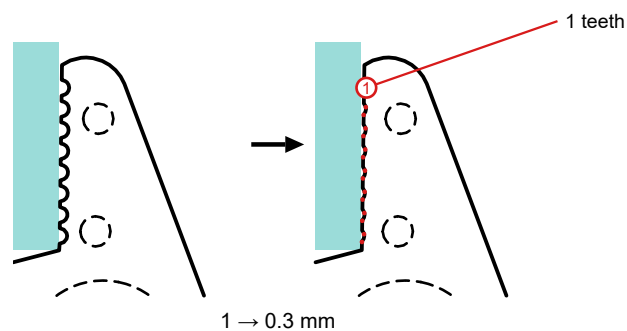
Default 2.8 · Range 0.5 to 5 · Step size 0.1 · Unit mm

Before: 2.8. After: 4.5.

Moves: teeth.

plate_tooth_depth

Tooth depth



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tooth depth at 0.3 mm: cuts each tooth deeper or shallower into the gripping edge. Marked in red: 1, the teeth: the serrated inner edges of the arms. The plate edge, the arms, the finger lobes and the cord channel stay where they were.

Cuts each tooth deeper or shallower into the gripping edge.

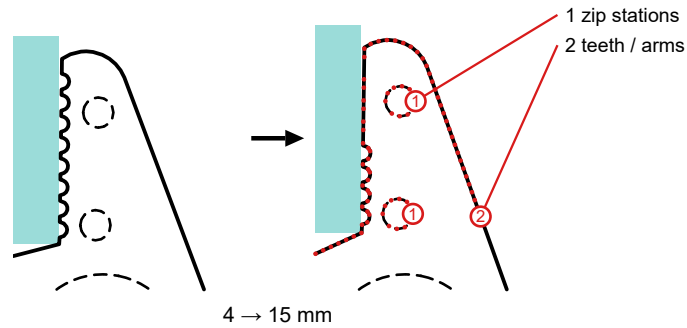
Default 1 · Range 0 to 1.5 · Step size 0.05 · Unit mm

Before: 1. After: 0.3.

Moves: teeth.

plate_grip_zone_start

Toothed zone start



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: toothed zone start at 15 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the teeth and the arms: the serrated inner edges of the arms; the two toothed arms in the upper half.

Moves where the teeth begin, measured back from the arm tips, and the arms lengthen when the zone needs the room.

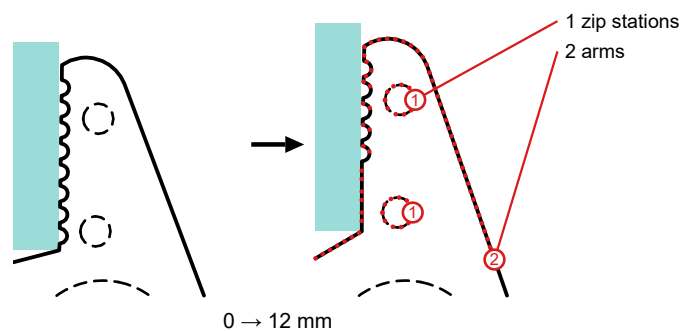
Default 4 · Range 0 to 25 · Step size 0.5 · Unit mm

Before: 4. After: 15.

Moves: arms, teeth, zip stations.

plate_grip_zone_length

Toothed zone length



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: toothed zone length at 12 mm: sets how far the teeth run along each arm instead of covering the whole plug body. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the arms: the two toothed arms in the upper half.

Sets how far the teeth run along each arm instead of covering the whole plug body.

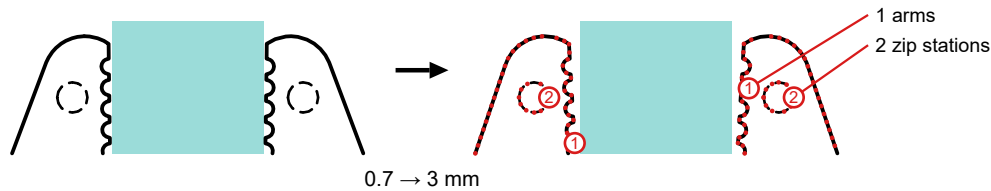
Default 0 · Range 0 to 60 · Step size 1 · Unit mm

Before: 0. After: 12.

Moves: arms, zip stations.

plate_tip_flare

Tip flare



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: tip flare at 3 mm: opens the gap wider right at the arm tips so the plug head can enter before the teeth bite. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm.

Opens the gap wider right at the arm tips so the plug head can enter before the teeth bite.

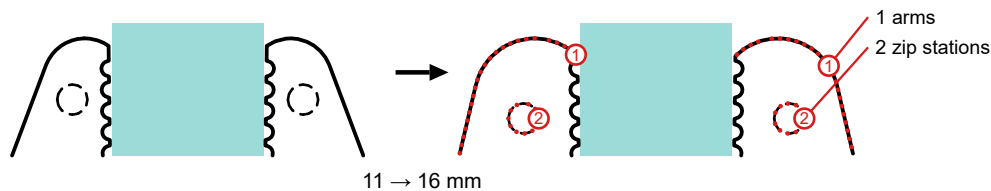
Default 0.7 · Range 0 to 4 · Step size 0.1 · Unit mm

Before: 0.7. After: 3.

Moves: arms, cord channel, finger lobes, zip stations.

plate_arm_tip_width

Arm tip width



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: arm tip width at 16 mm: widens the rounded tip of each arm, so the arms taper less. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm.

Widens the rounded tip of each arm, so the arms taper less.

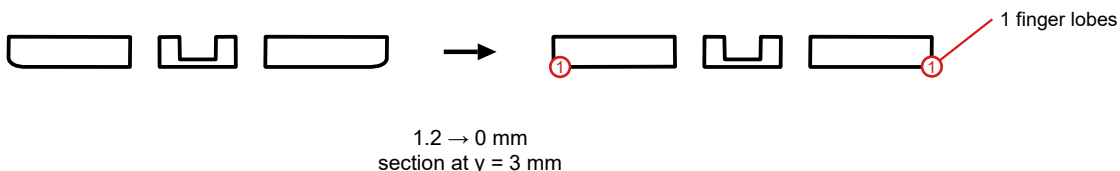
Default 11 · Range 5 to 16 · Step size 0.5 · Unit mm

Before: 11. After: 16.

Moves: arms, zip stations.

plate_edge_rounding

Outer edge rounding



Two vertical slices of the two-sided puller at y = 3 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: outer edge rounding at 0 mm: rounds or squares the outer face's edge all around the plate, while the plug-contact face stays square. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. The plate edge, the arms, the teeth and the cord channel stay where they were.

Rounds or squares the outer face's edge all around the plate, while the plug-contact face stays square.

Default 1.2 · Range 0 to 2 · Step size 0.1 · Unit mm

Before: 1.2. After: 0. Section at y = 3 mm.

Moves: finger lobes.

plate_strip_thickness

Cable strip thickness



1 → 0 mm
section at y = 3 mm

Two vertical slices of the two-sided puller at y = 3 mm, before left and after right, the top face up. Left: the defaults, the plug in teal in the cut. Right: cable strip thickness at 0 mm: thickens or removes the thin strip that bridges the cord channel on the outer face and ties the two arms together. Marked in red: 1, the cord channel: the gap between the lobes at the bottom. The plate edge, the arms, the teeth and the finger lobes stay where they were.

Thickens or removes the thin strip that bridges the cord channel on the outer face and ties the two arms together.

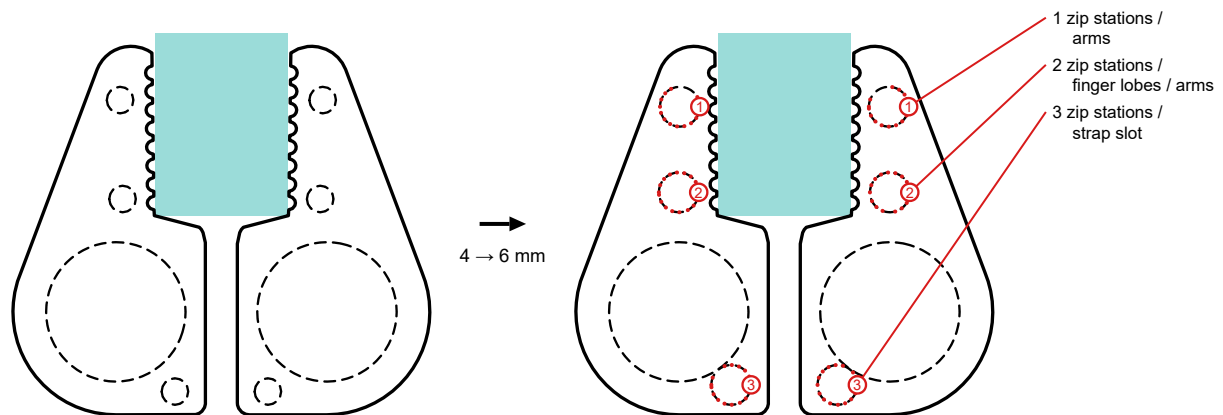
Default 1 · Range 0 to 4 · Step size 0.25 · Unit mm

Before: 1. After: 0. Section at y = 3 mm.

Moves: cord channel.

plate_zip_hole_diameter

Zip-tie hole diameter



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults, a 20 mm wide plug in teal. Right: zip-tie hole diameter at 6 mm. Marked in red: 1, the zip stations and the arms. 2, the zip stations, the finger lobes and the arms. 3, the zip stations and the strap slot.

Widens the three zip-tie holes in each arm, and the stations shift to keep their walls.

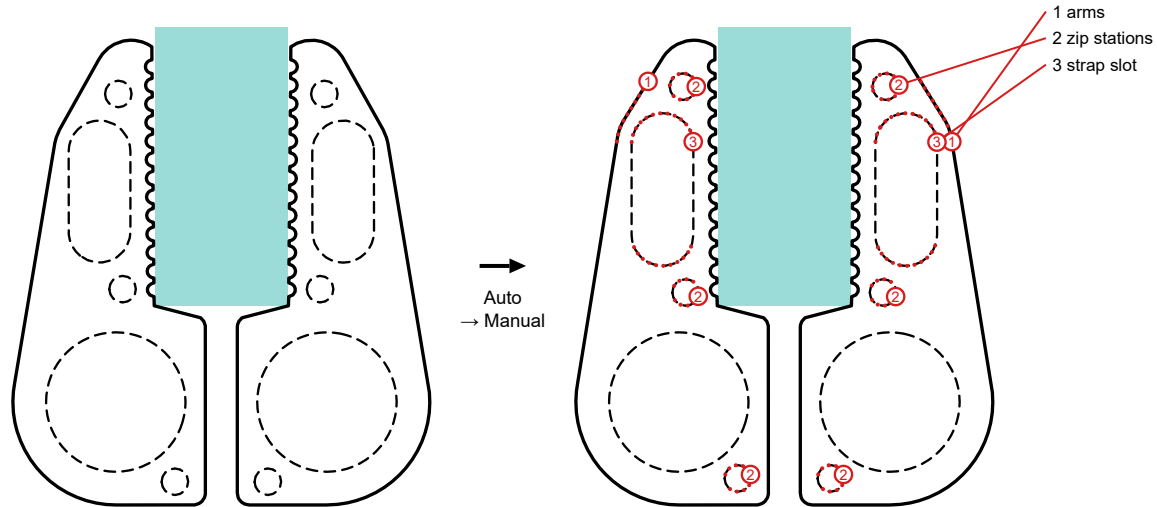
Default 4 · Range 0 to 8 · Step size 0.1 · Unit mm

Before: 4. After: 6.

Moves: arms, finger lobes, strap slot, zip stations.

plate_zip_placement

Zip-tie station placement



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: zip-tie station placement set to Manual. Marked in red: 1, the arms: the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm. 3, the strap slot: the long slot in each arm.

Switches the three zip-tie stations in each arm from the automatic positions to the three position dials.

Default Auto · 2 options

Options: Auto, Manual.

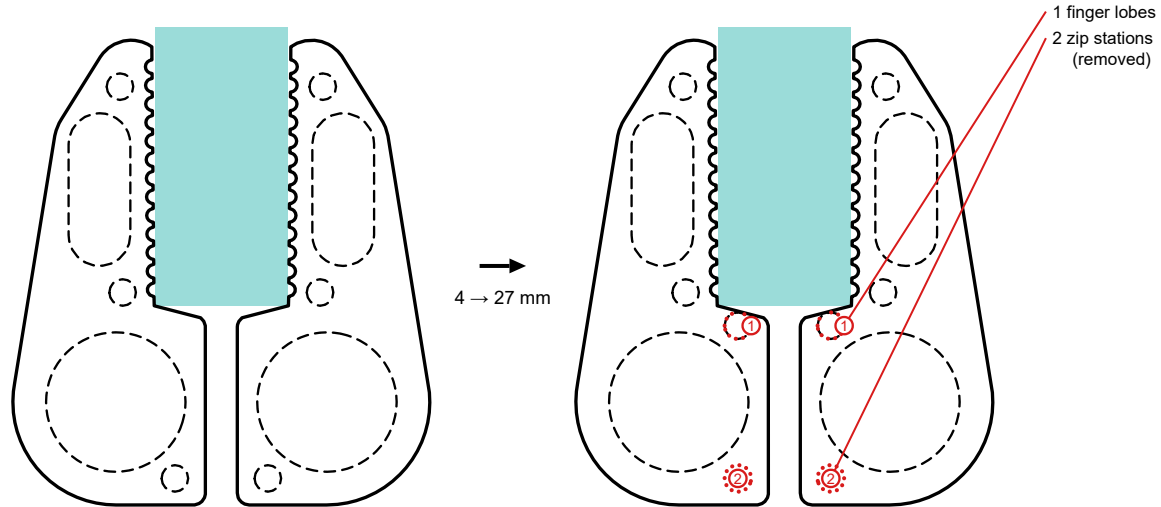
Before: Auto. After: Manual. Context: `measure_plug_length = 40`.

Drawn on a 40 mm plug so the default manual positions sit on the arm.

Moves: arms, strap slot, zip stations.

plate_zip_pos_1

Zip-tie station 1 position



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40 and zip-tie station placement set to Manual, a 20 mm wide plug in teal. Right: zip-tie station 1 position at 27 mm. Marked in red: 1, the finger lobes: the two rounded lobes in the lower half. 2, the zip stations: the three small holes along each arm, removed.

Slides the rear zip-tie station along each arm, measured from the cord end.

Default 4 · Range 0 to 80 · Step size 0.5 · Unit mm

Before: 4. After: 27. Context: measure_plug_length = 40, plate_zip_placement = Manual.

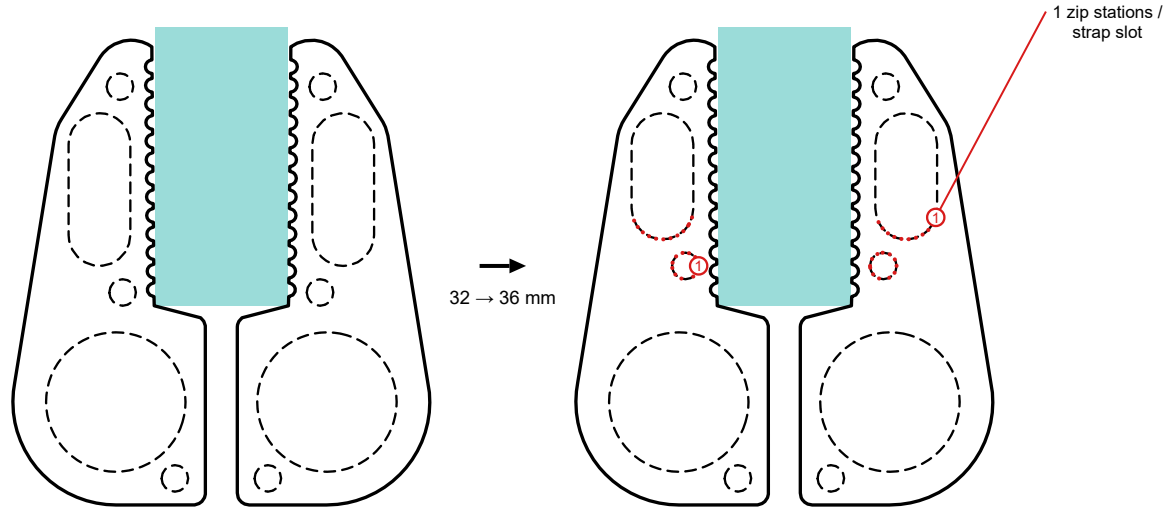
Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Moves: finger lobes, zip stations.

Can trip: ZIP STATION OFF THE ARM; ZIP STATIONS OVERLAP EACH OTHER; STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

plate_zip_pos_2

Zip-tie station 2 position



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40 and zip-tie station placement set to Manual, a 20 mm wide plug in teal. Right: zip-tie station 2 position at 36 mm. Marked in red: 1, the zip stations and the strap slot: the three small holes along each arm; the long slot in each arm.

Slides the middle zip-tie station along each arm, and the strap slot's window moves with it.

Default 32 · Range 0 to 80 · Step size 0.5 · Unit mm

Before: 32. After: 36. Context: measure_plug_length = 40, plate_zip_placement = Manual.

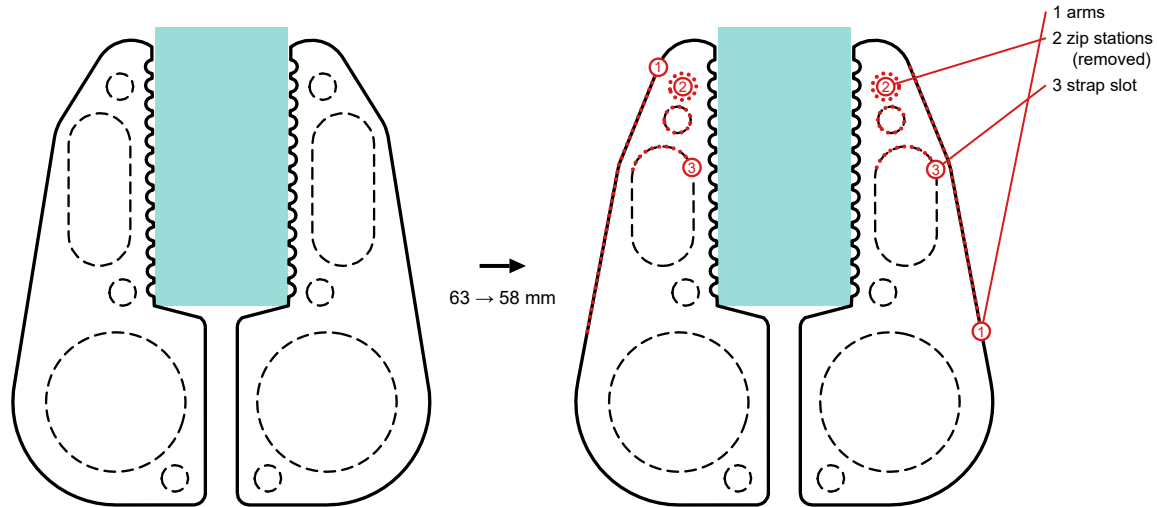
Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Moves: strap slot, zip stations.

Can trip: ZIP STATION OFF THE ARM; ZIP STATIONS OVERLAP EACH OTHER; ZIP STATION HITS VELCRO SLOT; STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

plate_zip_pos_3

Zip-tie station 3 position



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40 and zip-tie station placement set to Manual, a 20 mm wide plug in teal. Right: zip-tie station 3 position at 58 mm. Marked in red: 1, the arms; the two toothed arms in the upper half. 2, the zip stations: the three small holes along each arm, removed. 3, the strap slot: the long slot in each arm.

Slides the tip zip-tie station along each arm, and the strap slot's window moves with it.

Default 63 · Range 0 to 80 · Step size 0.5 · Unit mm

Before: 63. After: 58. Context: measure_plug_length = 40, plate_zip_placement = Manual.

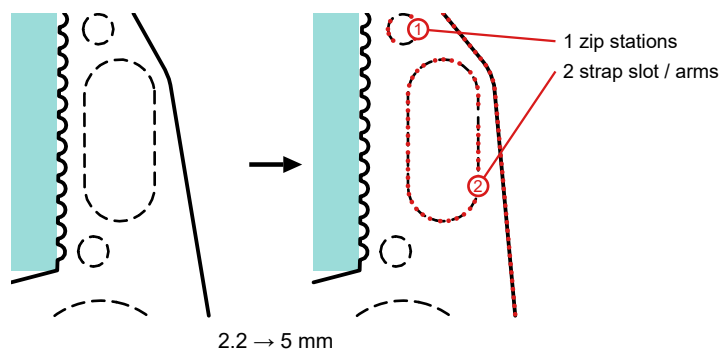
Only acts when plate_zip_placement is Manual; drawn on a 40 mm plug so every station sits on the arm.

Moves: arms, strap slot, zip stations.

Can trip: ZIP STATION OFF THE ARM; ZIP STATIONS OVERLAP EACH OTHER; ZIP STATION HITS VELCRO SLOT; STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

plate_slot_inner_wall

Wall between the teeth and the slot



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: wall between the teeth and the slot at 5 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Thickens the wall between the toothed edge and the strap slot, moving the slot outward.

Default 2.2 · Range 1 to 8 · Step size 0.1 · Unit mm

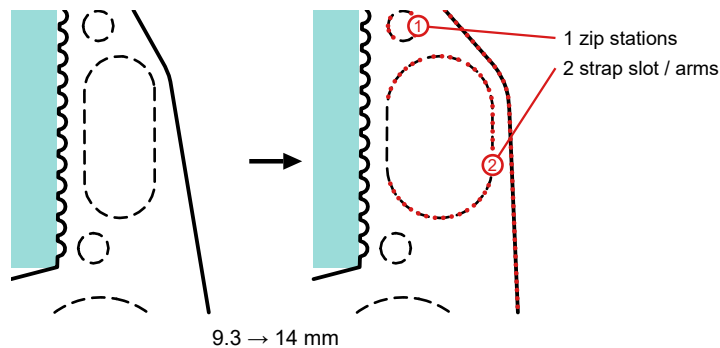
Before: 2.2. After: 5. Context: measure_plug_length = 40.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

plate_velcro_slot_width

Strap slot width



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap slot width at 14 mm. Marked in red: 1, the zip stations: the three small holes along each arm. 2, the strap slot and the arms: the long slot in each arm; the two toothed arms in the upper half.

Widens the strap slot in each arm, and the arm bulges outward to keep its wall.

Default 9.3 · Range 0 to 20 · Step size 0.25 · Unit mm

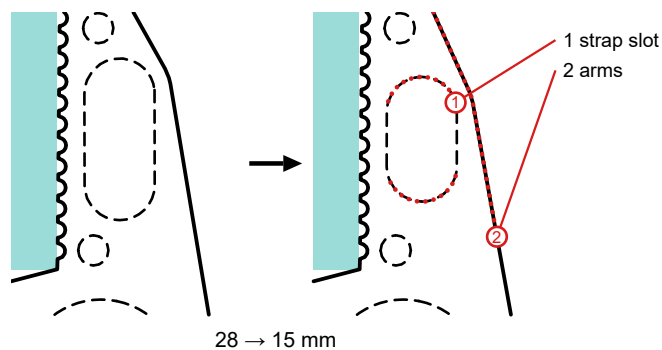
Before: 9.3. After: 14. Context: measure_plug_length = 40.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

Moves: arms, strap slot, zip stations.

plate_velcro_slot_length

Strap slot length



Two top views of one plate of the two-sided puller, before left and after right, the plug end at the top. Left: the defaults with plug length set to 40, a 20 mm wide plug in teal. Right: strap slot length at 15 mm: lengthens or shortens the strap slot along each arm, within the window between the zip-tie stations. Marked in red: 1, the strap slot: the long slot in each arm. 2, the arms: the two toothed arms in the upper half.

Lengthens or shortens the strap slot along each arm, within the window between the zip-tie stations.

Default 28 · Range 5 to 60 · Step size 1 · Unit mm

Before: 28. After: 15. Context: measure_plug_length = 40.

Drawn on a 40 mm plug: on the default 25.5 mm plug the arms are too short for a strap slot.

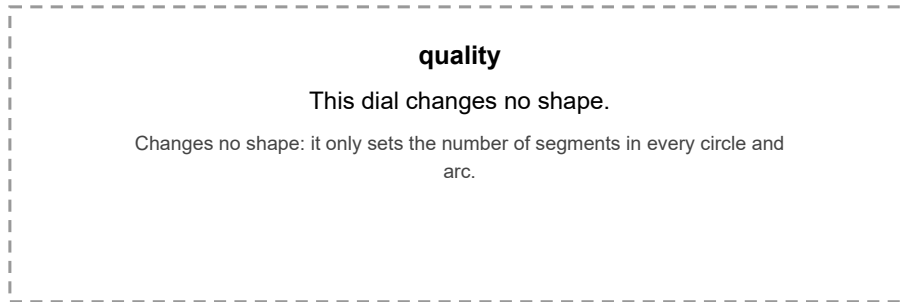
Moves: arms, strap slot, zip stations.

Can trip: STRAP WIDER THAN ARM SLOT WINDOW - NARROW THE STRAP.

Advanced - Render Quality

quality

Render quality



This dial changes no shape: it only sets the number of segments in every circle and arc.

Sets how many flat segments draw each curve and changes no dimension.

Default 64 · Range 24 to 128 · Step size 8 · Unit segments

Measuring guide

Everything here is in mm: a US plug is about 25 mm wide, so a 1 on your paper means you measured in inches. You need a caliper or a ruler with mm marks, the plug in its outlet, and your own hand only if you pick Measure my hand. Print the measuring form at 100 % and fill it in as you go: the sections below are the form's rows, in the same order, and the card names (R1, C1, F1 / F2) are the cards of the printed measuring stencil.

The two-sided puller's widths are the size the two plates close across, so measure across the plug the way the plates will grip it. The thickness pair, the wall plate style and the hand width belong to the one-sided puller only.

2. Plug length

Customizer name `measure_plug_length`, row 2 of the form.

With the plug in the outlet, ruler from the wall plate face to the plug's back face. On a laptop or charger plug, measure from the device's edge.

Typical: 20 to 50 mm. Example: 23 mm.

With the stencil: card R1.

3. Plug width at the prong end

Customizer name `measure_plug_width_prong_end`, row 3 of the form.

Caliper across the plug body just behind the prongs, the size the two plates close across. On a USB-C or charger tip, just behind the metal tip; on a round plug it is the diameter.

Typical: 12 to 40 mm. Example: 13 mm.

With the stencil: card R1.

4. Plug width at the cord end

Customizer name `measure_plug_width_cord_end`, row 4 of the form.

The same direction across the plates, measured where the cord leaves the plug body. Skip the soft rubber strain relief.

Typical: 12 to 40 mm. Example: 13 mm.

With the stencil: card R1.

5. Cord thickness

Customizer name `measure_cord_thickness`, row 5 of the form.

Caliper across the cord just behind the plug, on its thin side: flat cord the narrow way, round cord the diameter. With the stencil, the smallest C1 slot that slips over the cord is this number.

Typical: 3 to 9 mm. Example: 7 mm.

With the stencil: card C1.

6. Plug sides

Customizer name `plug_sides`, row 6 of the form.

Look at the plug body where the plates will grip it. Rounded sides for a round cord plug, a USB-C or charger tip; Flat sides for a boxy plug.

The choices: Rounded sides, Flat sides.

9. Finger width

Customizer name `measure_finger_width`, row 9 of the form.

Caliper across the widest knuckle of your middle finger, the finger that goes into the pull hole. With the stencil, find the smallest F1 or F2 hole your finger passes through comfortably and subtract 5 from its number.

Typical: 16 to 24 mm. Example: 22 mm.

With the stencil: card F1 / F2.

No caliper? Take a ring that fits that finger snugly, measure the ring's inner diameter in mm and add 1.5 mm.

11. Strap width

Customizer name `strap_width`, row 11 of the form.

Read the width on the strap's packaging. ONE-WRAP comes in 10, 13, 16, 20 and 25 mm.

Typical: 10 to 25 mm. Example: 15 mm.

Sanity check: each plug width is a two-digit number, roughly 12 to 45, and the prong-end width is usually the bigger one; the finger width is roughly 14 to 32. A number like 1.3 is inches: measure again with the mm side.

When you measure for someone else, a relative or a client, measure their hand for the finger and hand rows and their outlet and plug for the plug rows. Doubtful between two values? Round up: a slightly roomy fit works, a tight one does not.

Measuring form

The next page is the measuring form: print it at 100 % (actual size, never fit to page) and check the 50 mm bar with a ruler before you trust it. Fill in the blanks top to bottom, then type the numbers into the Customizer in the same order. Its text twin is `measuring-form.md` beside this packet's twins.

Plug Puller — Measuring form, two-sided puller

Measure in mm. Fill in the blanks top to bottom, then type them into the Customizer in the same order.

Print at 100 % 50 mm

- ① **plug_preset**
Plug preset
☒ **Measure my plug**
☐ Heavy-duty extension cord - NEMA 5-15
☐ USB-C laptop tip ☐ Flat 2-prong lamp plug - NEMA 1-15
☐ Standard 3-prong plug - NEMA 5-15
- ② **measure_plug_length**
Plug length
default 25.5 mm → _____ mm card R1
- ③ **measure_plug_width_prong_end**
Plug width at the prong end
default 20 mm → _____ mm card R1
- ④ **measure_plug_width_cord_end**
Plug width at the cord end
default 20 mm → _____ mm card R1
- ⑤ **measure_cord_thickness**
Cord thickness
default 4 mm → _____ mm card C1
- ⑥ **plug_sides**
Plug sides
☒ **Rounded sides** ☐ Flat sides
- ⑦ **show_plug_preview**
Show the see-through plug
☒ **leave on**
- ⑧ **size**
Hand size
☐ Small ☒ **Medium** ☐ Large ☐ Measure my hand
- ⑨ **measure_finger_width**
Finger width
default 20 mm → _____ mm card F1 / F2
- ⑩ **attachment**
Attachment
☒ **Zip ties + Velcro strap** ☐ Zip ties
- ⑪ **strap_width**
Strap width
default 15 mm → _____ mm
- ⑫ **print_layout**
Print layout
☒ **Both plates** ☐ One plate

